TECHNICAL DOCUMENTATION v1.0

Her Wellness

PCOS & PCOD Companion

System Design, Architecture & Build Documentation

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1 Executive Overview

Purpose, Goals & Product Summary

Her Wellness is a comprehensive, AI-powered web application built to support women living with Polycystic Ovarian Syndrome (PCOS) and Polycystic Ovarian Disease (PCOD). The application delivers personalised health assessments, evidence-based care guidance, educational content, cycle tracking, mental wellness support, and real-time AI-driven Q&A; — all within a single, accessible interface deployed as a claude.ai Artifact.

Goal	Description	Priority
Personalised Assessment	9-question clinical-style intake to profile symptoms, lifestyle, and goals	`P0 — Core`
Education & Awareness	Evidence-based content covering PCOS vs PCOD, causes, types, tests, and symptoms	`P0 — Core`
Care Plan Delivery	Structured guidance across diet, exercise, sleep, supplements, herbs, and medical options	`P0 — Core`
Menstrual Cycle Tracking	Interactive calendar for period/ovulation marking and daily symptom logging	`P1 — High`
Mental Wellness	Affirmations, coping tools, body-positive content, and crisis helplines (India)	`P1 — High`
AI-Powered Q&A;	Real-time conversational AI using Claude Sonnet 4.6 with PCOS-specialised system prompt	`P0 — Core`

Target Audience

Women aged 15–50 in India experiencing or managing PCOS/PCOD symptoms. The application is designed for self-education and symptom awareness — it supplements, but does not replace, professional medical advice from a certified gynecologist or endocrinologist.

2 System Architecture

Layered Architecture & Component Design

2.1 Architecture Diagram — Layered View

The application follows a five-tier client-side architecture. All business logic, state management, and UI rendering execute entirely within the browser sandbox. External communication is limited to the Anthropic API endpoint for AI Q&A; functionality.

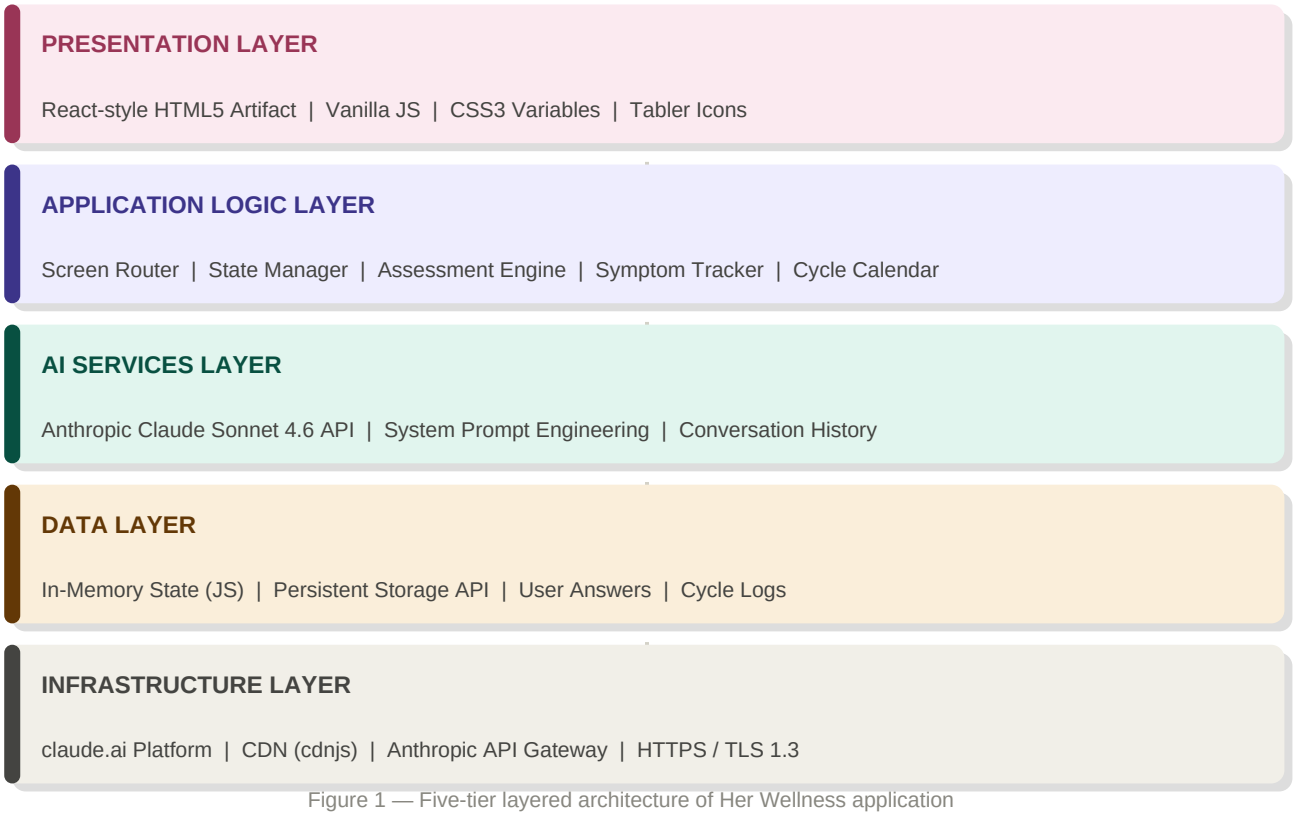

Figure 1 — Five-tier layered architecture of Her Wellness application

2.2 Component Architecture

Six primary functional modules compose the application. Each module is self-contained with its own view state, data bindings, and event handlers, orchestrated by a lightweight screen-routing system.

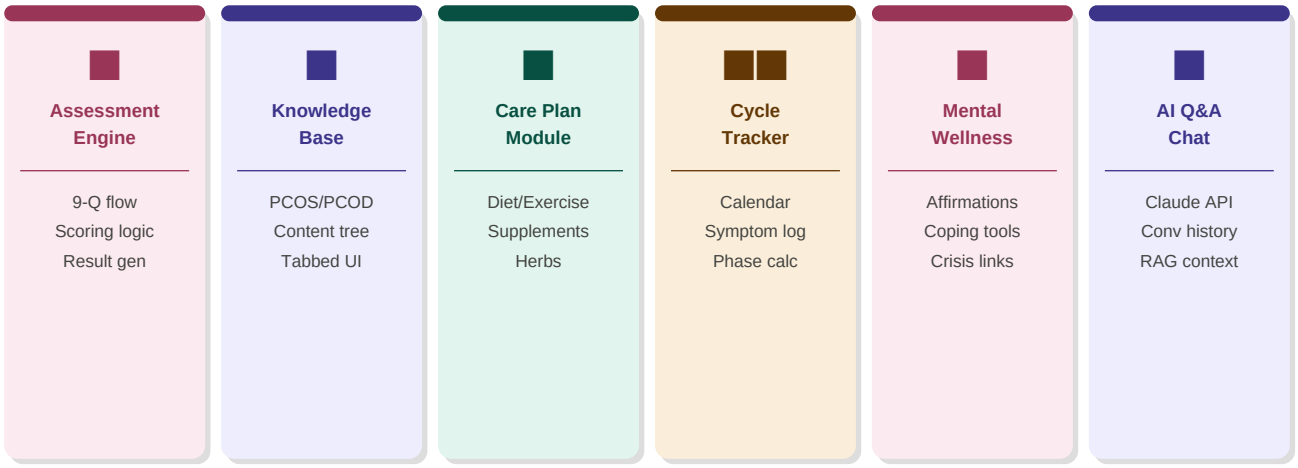

Figure 2 — Six core functional modules

2.3 Data Flow Diagram

The following table describes all primary data flows within the application, from user interaction through processing to output.

Flow ID	Source	Destination	Data	Trigger
DF-01	User Input	Assessment Engine	Answer selection (string/array)	onClick
DF-02	Assessment Engine	Result Generator	answers{} object — 9 keys	nextQuestion()
DF-03	Result Generator	Result Screen	Severity score, top tips, goal advice	showResults()
DF-04	User Input	AI Q&A; Module	Natural language question string	askQuestion()
DF-05	AI Q&A; Module	Anthropic API	messages[] + system prompt + model config	fetch() POST
DF-06	Anthropic API	AI Q&A; Module	Claude response (content[].text)	async response
DF-07	AI Q&A; Module	Conversation History	Updated messages[] array	After each turn
DF-08	User Input	Cycle Tracker	Day number + mode (period/ovulation)	onClick calendar
DF-09	User Input	Symptom Log	Selected symptom tags array	toggleSymptom()
DF-10	Storage API	All Modules	Persisted state JSON from previous sessions	On load

3 Module Specifications

Detailed design of each functional module

3.1 Assessment Engine

The Assessment Engine is the diagnostic intake module. It presents a sequential 9-question survey covering demographics, menstrual health, physical symptoms, mental symptoms, diagnosis status, lifestyle, diet, primary goal, and emotional state.

Key Functions:

Function	Description
`initAssessment()`	Resets state, clears answers{}, renders Q1
`renderQuestion()`	Injects HTML for current question — progress bar, prompt, options grid, Next button
`selectOpt(el, key, val)`	Single-select: clears all opts, marks selected, stores answers[key]=val
`toggleOpt(el, key, val)`	Multi-select: toggles class, pushes/splices answers[key][] array
`nextQuestion()`	Increments currentQ, calls renderQuestion() or showResults()
`showResults()`	Calculates severity score, severity class, detects irregularity, renders personalised result screen
`getGoalAdvice(goal)`	Returns goal-specific HTML action plan from 7 keyed templates

Scoring Algorithm

Severity is computed by summing all selected symptom tags from questions 3 and 4 (max 12). Score 0–3 = Mild (green), 4–7 = Moderate (amber), 8–12 = Significant (red). Cycle regularity is assessed from Q2 response. Irregular + Significant triggers a "see a doctor" alert. Goal-specific advice is keyed to Q8 response — 7 distinct care plans.

3.2 Knowledge Base Module

Delivers static educational content across 5 topics via a tab navigation pattern. Content is authored directly in JavaScript template literals — no external CMS or API required. The switchTab() function manages active class toggling between tab buttons and content blocks.

Tab ID	Title	Content
`l-diff`	PCOS vs PCOD	Side-by-side comparison cards — disease vs syndrome, severity, reversibility
`l-causes`	Causes	6 root causes: genetics, insulin resistance, inflammation, androgens, lifestyle, stress
`l-types`	Types of PCOS	4 clinical types: insulin-resistant, inflammatory/adrenal, post-pill, hidden/lean
`l-sympt`	Symptoms	Tagged symptom groups: common, metabolic, reproductive — 18 total symptoms

`l-tests`	Tests & Diagnosis	Rotterdam Criteria explanation, 5 test categories, what to ask your doctor

3.3 Daily Care Plan Module

Six-tab interface delivering actionable daily care guidance. Each tab is a self-contained educational section with formatted tip lists, food grids, and structured recommendations.

Tab	Key Content	Format
Diet	Eat/avoid food lists (Indian-friendly), low-GI principles, meal timing, spearmint tea	`2-col food grid + tip list`
Exercise	Walking, strength training, yoga poses, cardio, over-exercise warning	`Icon-led tip list`
Sleep	7–9hr target, schedule consistency, screen-free bedtime, room conditions, caffeine	`Icon-led tip list`
Supplements	Inositol, Vit D3+K2, Omega-3, Magnesium, NAC, Iron/B12 — with doses	`Icon-led tip list + warning`
Herbs	Spearmint, Shatavari, Ashwagandha, Chasteberry, Cinnamon, Fenugreek	`Icon-led tip list`
Medical	Metformin, OCP, Spironolactone, Letrozole, monitoring schedule	`Icon-led tip list + warning`

3.4 Cycle Tracker Module

Interactive menstrual cycle tracker with three subcomponents: educational phase diagram, interactive monthly calendar, and daily symptom log.

- Phase Diagram: Visual bar showing menstruation (5d), follicular (9d), ovulation (1d), luteal (14d) phases with colour-coded legend
- Calendar Grid: CSS grid of 30 day-cells for June 2026. Click toggles between period (pink) / ovulation (teal) / unset states. State stored in markedDays{} object keyed by day number
- Mode Toggle: setCycleMode() switches between 'period' and 'ovulation' marking modes with visual label feedback
- Symptom Log: 9 symptom chips (cramps, fatigue, mood, headache, bloating, cravings, insomnia, brain fog, anxiety). Multi-select with toggleSymptom(). saveSymptoms() shows a 3-second success toast

3.5 Mental Wellness Module

Provides emotional support, psychoeducation, and coping tools. The affirmation rotator cycles through 8 curated body-positive statements via nextAffirmation(). India-specific crisis helplines (iCall: 9152987821, Vandrevala: 1860-2662-345) are always visible.

3.6 AI Q&A; Chat Module

The AI module integrates directly with the Anthropic Messages API to deliver real-time, PCOS-specialised conversational responses. The module maintains full conversation history across the session for contextual multi-turn dialogue.

System Prompt Design:

```
You are a warm, knowledgeable, and empathetic PCOS & PCOD health companion for women in India. Always respond with: clear, compassionate, non-judgmental language | practical, actionable advice for India | evidence-based information | emotional validation | a reminder to consult a gynecologist | 200–400 words | empowering language | cultural sensitivity (Indian foods, herbs, accessibility). Never provide a specific diagnosis or prescribe medications.
```

Quick Suggestion Chips:

- Can I get pregnant with PCOS? — triggers fillQuestion() to pre-fill the textarea
- What foods help reduce PCOS? — triggers fillQuestion() to pre-fill the textarea
- How do I know if I have PCOS or PCOD? — triggers fillQuestion() to pre-fill the textarea
- What exercises are best for PCOS? — triggers fillQuestion() to pre-fill the textarea

4 Technology Stack

All technologies, libraries, and dependencies

Category	Technology	Version	Purpose
Runtime	**Web Browser**	`Modern (ES2020+)`	Application host — Chrome, Safari, Firefox, Edge
Markup	**HTML5**	`5`	Semantic structure, accessibility, ARIA
Styling	**CSS3**	`3`	Layout, theming, animations, responsive design
Scripting	**Vanilla JavaScript**	`ES2020`	All application logic, DOM manipulation, API calls
CSS System	**claude.ai CSS Variables**	`Platform`	Colour theming, dark/light mode, typography
Icons	**Tabler Icons (outline)**	`v3.x webfont`	UI iconography — outline style, 5800+ icons
Fonts	**Anthropic Sans**	`Platform`	Primary typeface via --font-sans variable
AI Service	**Anthropic Claude Sonnet**	`4.6`	AI Q&A; — claude-sonnet-4-6 model
AI Protocol	**Anthropic Messages API**	`v1`	POST /v1/messages — streaming-capable
Storage	**window.storage API**	`Platform`	Persistent key-value store across sessions
Platform	**claude.ai Artifact Env**	`Current`	Sandboxed iframe rendering environment
CDN	**cdnjs.cloudflare.com**	—	External library delivery (CSP-approved)

5 AI Integration Design

Anthropic API integration architecture

API Request Configuration:

Parameter	Value
Endpoint	`https://api.anthropic.com/v1/messages`
Method	`POST`
Model	`claude-sonnet-4-6`
Max Tokens	`1000 per response`
Auth	`Handled by claude.ai platform (no API key in client code)`
Content-Type	`application/json`

Conversation Management:

The convHistory[] array persists throughout the session, storing alternating user/assistant message objects. Each API call sends the full history ensuring contextual, coherent multi-turn responses. The array is initialised empty on page load and grows with each exchange.

```
const convHistory = [] convHistory.push({ role: 'user', content: question }) // API call with full history... convHistory.push({ role: 'assistant', content: reply })
```

Error Handling

The askQuestion() function wraps the fetch() call in a try/catch block. On network error or API failure, a user-friendly error message is rendered in the response container directing the user to the static care guides as a fallback. Loading state is indicated by three animated dots while the API request is in flight.

6 State Management

In-memory state architecture

Variable	Init	Module	Purpose
`answers`	`{}`	Assessment Engine	Stores user responses keyed by question key
`currentQ`	`0`	Assessment Engine	Current question index (0–8)
`convHistory`	`[]`	AI Q&A; Module	Full message history for Anthropic API
`cycleMode`	`'period'`	Cycle Tracker	Current calendar marking mode
`markedDays`	`{}`	Cycle Tracker	Day-number keyed calendar state object
`affirmIdx`	`0`	Mental Wellness	Current affirmation index (0–7)
`affirmations`	`Array[8]`	Mental Wellness	Static array of affirmation objects {icon, text}
`questions`	`Array[9]`	Assessment Engine	Static question definition array

Persistent Storage

The window.storage API (provided by claude.ai) enables cross-session persistence. The API supports get(key), set(key, value), delete(key), and list(prefix) with optional shared mode. All storage operations are async and wrapped in try/catch. Personal data is scoped to the current user (shared=false default). Storage is used for cycle calendar state, symptom logs, and assessment history.

7 UI/UX Design System

Visual design tokens, typography, and patterns

Colour Palette:

Name	Hex	Role	Usage
Pink Dark	`#993556`	**Primary Brand**	CTA buttons, active states, links
Pink Mid	`#D4537E`	**Primary Accent**	Selected elements, borders, icons
Pink Light	`#FBEAF0`	**Primary Background**	Card backgrounds, tag fills, hero bg
Purple Dark	`#3C3489`	**Secondary Brand**	Section headers, headings, accent text
Purple Mid	`#7F77DD`	**Secondary Accent**	AI response border, tab active (alt)
Purple Light	`#EEEDFE`	**Secondary Background**	Module backgrounds, learn cards
Teal Dark	`#085041`	**Success/Positive**	Positive indicators, eat-more foods
Teal Mid	`#1D9E75`	**Success Accent**	Ovulation marker, positive badges
Teal Light	`#E1F5EE`	**Success Background**	Positive tip boxes, teal card fills
Amber Mid	`#BA7517`	**Warning**	Caution notices, warnings
Amber Light	`#FAEEDA`	**Warning Background**	Warning boxes, disclaimer backgrounds

Typography Scale:

- H1 — 22px / weight 500 / Anthropic Sans — Page section titles
- H2 — 18px / weight 500 / Anthropic Sans — Sub-section headings
- H3 — 16px / weight 500 / Anthropic Sans — Card headings
- Body — 16px / weight 400 / Anthropic Sans / line-height 1.7 — Primary prose
- Caption — 13px / weight 400 / GRAY_500 — Labels, metadata, helper text
- Monospace / Code — Courier / 8.5–14px / PURPLE_DARK — Code, function names

8 Security & Privacy

Data handling, sandbox constraints, and safety

No API Key Exposure: The Anthropic API key is never present in client-side code. Authentication is handled transparently by the claude.ai platform infrastructure.

Sandbox Isolation: The application runs inside a sandboxed iframe with Content Security Policy enforced. External resources are restricted to a CDN allowlist (cdnjs.cloudflare.com, esm.sh, cdn.jsdelivr.net, unpkg.com).

No PII Transmission: User health data (symptoms, cycle data, answers) is stored only in window.storage (client-side) and is never transmitted to any third-party server.

No localStorage: The application uses window.storage (platform API) rather than browser localStorage/sessionStorage, which are unsupported in the claude.ai artifact environment.

Medical Disclaimer: A persistent disclaimer is shown on the home screen and assessment results page. The AI system prompt explicitly prohibits diagnosis and prescriptions.

Content Safety: The AI system prompt includes guardrails against providing specific medication doses, direct medical diagnosis, or advice that could replace professional care.

Crisis Resources: India-specific mental health helplines (iCall, Vandrevala Foundation) are hardcoded in the Mental Wellness module and always visible.

9 Performance Specifications

Load targets and optimisation strategies

Metric	Target	Strategy
Initial Render	**< 500ms**	Inline CSS/JS — no external stylesheets to block rendering
Screen Transition	**< 50ms**	Pure CSS display toggle — no DOM re-creation between screens
AI Response	**< 3s**	Claude Sonnet 4.6 API — fast inference model; 1000 max_tokens limit
Calendar Render	**< 10ms**	Pure DOM manipulation — 30 cell insertions, no framework overhead
Bundle Size	**< 60KB**	Zero dependencies — vanilla HTML/CSS/JS only (no React, no build step)
CSS Variables	**Real-time**	System variable resolution handled natively by browser — zero cost
Dark Mode Switch	**< 16ms**	CSS variable cascade — single class toggle on :root propagates all colours

10 Build & Deployment Guide

How to run, modify, and deploy the application

10.1 Prerequisites

- A claude.ai account (Free, Pro, or Team tier)
- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- No local build tools, Node.js, or package managers required — zero build step
- No API key management — authentication handled by the claude.ai platform

10.2 Deployment Steps

1. **Open claude.ai**
   Navigate to claude.ai and log in with your account
2. **Start a new conversation**
   Click 'New chat' from the sidebar
3. **Request artifact creation**
   Paste the full application source code into the message box preceded by: 'Create an artifact with this code:'
4. **Artifact renders**
   The application renders inline in the conversation as an interactive artifact — no additional configuration needed
5. **Share / embed**
   Use the share button on the artifact to generate a shareable link, or embed via claude.ai's sharing features

10.3 Customisation Guide

Add/Edit Questions: Modify the questions[] array in the JavaScript section. Each question object requires: q (text), sub (subtext), opts (array), key (storage key), multi (boolean)

Update Content: Care plan, Knowledge Base, and Mental Wellness content is authored directly in template literals within the showScreen() calls. Search for the relevant HTML block by tab ID.

Change AI Behaviour: Edit the system prompt string inside the askQuestion() fetch body. The system prompt controls tone, scope, and safety guardrails.

Add Affirmations: Extend the affirmations[] array with new {icon, text} objects. nextAffirmation() cycles through the full array automatically.

Modify Colours: The CSS style block defines all custom class colours (PINK_*, PURPLE_*, TEAL_*). System colours use --color-* CSS variables from the claude.ai design system.

Add New Screens: Create a new div element with class='screen' and a unique ID, add a .home-card button pointing to it, and call showScreen('screen-xxx') from the routing function.

11 API Reference

All public functions and their signatures

Screen Router

Signature	Description
`showScreen(id: string)`	Hides all .screen elements, shows the target screen. Calls initAssessment() for assess screen, initCalendar() for cycle screen.

Assessment Engine

Signature	Description
`initAssessment()`	Resets currentQ=0, answers={}, calls renderQuestion()
`renderQuestion()`	Injects question HTML based on currentQ. Calls showResults() when currentQ reaches questions.length
`selectOpt(el, key, val)`	Single-select handler. Sets answers[key]=val, enables Next button
`toggleOpt(el, key, val)`	Multi-select handler. Toggles answers[key][] array membership
`nextQuestion()`	Increments currentQ, calls renderQuestion()
`showResults()`	Computes severity, formats result screen, calls showScreen('screen-result')
`getGoalAdvice(goal: string)`	Returns goal-keyed HTML string for 7 possible primary goals

Tab Navigation

Signature	Description
`switchTab(el, group, contentId)`	Deactivates all .tab in parent, activates el. Deactivates all .content-block, activates target

Cycle Tracker

Signature	Description
`initCalendar()`	Renders 30-cell calendar grid with correct first-day offset. Re-reads markedDays{} for colouring
`setCycleMode(mode: string)`	Sets cycleMode to 'period' or 'ovulation', updates mode label UI
`toggleSymptom(el)`	Toggles .selected class on symptom chip element
`saveSymptoms()`	Shows success toast for 3000ms via setTimeout

Mental Wellness

Signature	Description
`nextAffirmation()`	Increments affirmIdx (wraps at length), updates icon and text elements

AI Q&A;

Signature	Description
`fillQuestion(text: string)`	Sets textarea value to pre-defined question string
`askQuestion()`	Reads textarea, pushes to convHistory, calls Anthropic API, renders response

12 Testing Strategy

Quality assurance and validation approach

Test Type	Scope	Method	Priority
Functional	All 6 screen modules, navigation, all tabs	Manual browser testing — complete user journeys	**P0**
Assessment Flow	9-question flow with all option combinations	Manual — test single-select, multi-select, all goals	**P0**
AI Integration	Q&A; module — various question types and lengths	Live API testing with suggested chips + free-form	**P0**
Responsive Design	Mobile (375px), tablet (768px), desktop (1200px)	Browser DevTools device emulation	**P1**
Dark Mode	All screens in light and dark mode	Toggle OS dark mode — verify all text contrast	**P1**
Accessibility	Screen reader, keyboard navigation, ARIA labels	Axe DevTools audit, VoiceOver/NVDA manual test	**P1**
Performance	Load time, render performance, API latency	Chrome DevTools Lighthouse audit — target >90 score	**P2**
Content Accuracy	All medical/nutritional/clinical content	Review by qualified gynecologist or PCOS specialist	**P0**
Storage	window.storage persistence across sessions	Manual — mark calendar, reload page, verify state	**P2**

13 Roadmap & Future Features

Planned enhancements and v2.0 scope

Feature	Description	Version	Effort
Multi-Month Tracker	Rolling 6-month calendar with trend visualisation and cycle length stats	**v1.1**	Medium
Doctor Report Export	PDF symptom report auto-generated for doctor appointments	**v1.1**	Medium
Personalised Meal Plans	AI-generated weekly Indian meal plans based on PCOS type and dietary preference	**v1.2**	High
Push Notifications	Period reminders, medication reminders, water intake alerts via PWA	**v1.2**	High
Lab Results Interpreter	Upload blood test PDF; AI extracts and explains hormonal markers in plain language	**v2.0**	High
PCOS Type Identifier	Advanced assessment to determine which of 4 PCOS types is most likely	**v2.0**	Medium
Community Forum	Moderated peer support forum — shared storage for community Q&A;	**v2.0**	High
Yoga Video Library	Curated video playlist integration for PCOS-specific yoga routines	**v2.0**	Medium
Hindi / Regional Lang	Multilingual support: Hindi, Tamil, Telugu, Marathi, Bengali	**v2.0**	High
Wearable Integration	Import data from Google Fit / Apple Health for automated cycle and activity logging	**v3.0**	Very High
Telehealth Booking	In-app appointment booking with PCOS-specialist gynecologists in India	**v3.0**	Very High
Fertility Predictor	Ovulation prediction using BBT + cycle data + AI pattern recognition	**v3.0**	High

A Note on Mission

Her Wellness was built with one purpose: to ensure no woman feels alone or uninformed on her PCOS/PCOD journey. 1 in 5 women in India has PCOS, yet 70% are undiagnosed. This application is a step toward closing that gap — providing accessible, evidence-based, compassionate health support to every woman, wherever she is.